Imperial College London

Department of Earth Science and Engineering

MSc in Applied Computational Science and Engineering

Independent Research Project
Project Plan

# Learning Temporally Abstract MDPs

by

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# Abstract

One of the main challenges in reinforcement learning is to enable models to imagine trajectories over long horizons, which is crucial for creating agents capable of dealing with complex environments like NetHack and Minecraft. Current methods use world models which describe each state transition, making reasoning over longer horizons challenging. We plan to use the Predictron architecture, which was shown to exhibit abstraction capabilities in Markov Reward Processes, to learn a temporally abstract Markov Decision Process that approximates the original environment dynamics. We hope to show that making the world model reason at a more abstract level will facilitate thinking over longer horizons. We plan on testing our results on a variety of common benchmarks like MiniGrid and MiniHack.


## 1 Introduction

Reinforcement learning (RL) has been successfully used to solve many complex tasks ranging from playing Atari [14] to complex board games like Go [21] and Chess [22]. Many of these problems require a strong ability to think about the future. Various frameworks have emerged to do this, many of which rely on the idea of learning a world model. However, current methods have a hard time thinking over long horizons, and tasks like playing NetHack [12] and Minecraft [11] remain unsolved.

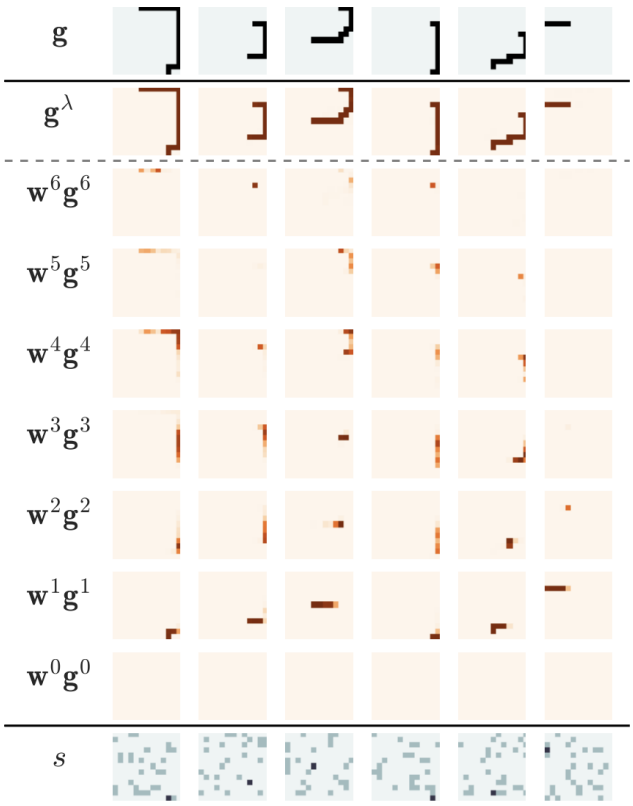
World models [5] are typically studied in model-based RL. They have many different benefits like enabling planning [21], training policies in imagination [5, 8] and differentiating through the learned dynamics [9]. However, current methods still struggle to learn world models that are able to imagine trajectories over long horizons [19, 10]. One reason for this limitation is that current methods try to reproduce every state transition of the original Markov Decision Process (MDP). This results in a very complex problem to solve, especially with high dimensional observations (e.g. pixels), as it requires capturing very fine-grained information to predict the next state. As a result, current methods require strong learning signals [10] and have difficulties in imagining trajectories over long horizons.

The research question we are interested in with this project is whether it is possible to construct a temporally abstract MDP to plan over longer horizons and automatically discover skills (abstract states). This would mean predicting the dynamics of the original MDP at a granular level. Our intuition is that such an MDP would not require fine-grained information as normal methods do, enabling it to work with weaker learning signals. As an example, imagine how a traditional world model would represent a person going to work: it would predict every frame, from the moment the person exits their home until they arrive at work. On the other hand, consider how a model capable of temporal abstraction would represent the same situation: it would think about opening the door, walking to the tube, and so on, without thinking about every specific step taken in the process. This is a much easier task than predicting everything that will happen frame-by-frame. However, these abstract states are generally not given and must be learnt. Our question then is: is it possible to learn these abstract states by learning a temporally-abstract MDP approximating the original one? Can we use these states for planning?

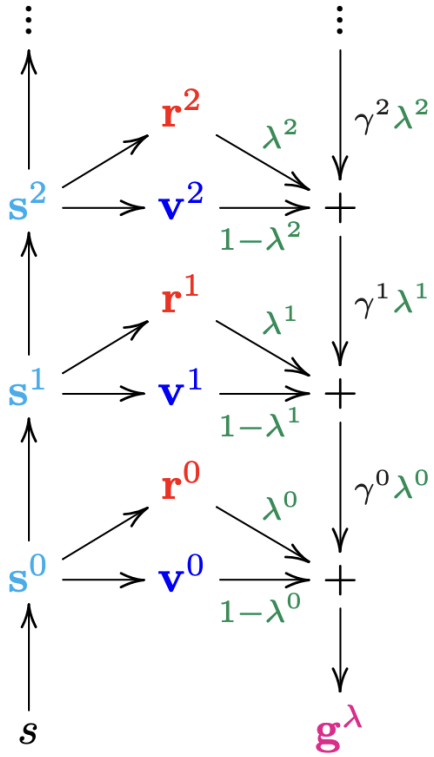
## 2 Methodology

### 2.1 Temporal Abstraction in MRPs

Tamar et al., 2016 [25] noted that model-free methods plan implicitly through the Bellman updates and proposed a new architecture (VIN) that incorporates planning within the neural network itself by using combinations of convolutions and max-pooling operations to perform the Bellman updates. Silver et al., 2017 [20] built on the ideas of VIN and extended them to more general non-grid world cases. They show that, using their Predictron architecture to model Markov Reward Processes (MRPs), they manage to get sequential temporal abstraction as the algorithm learns to solve various environments. The Predictron architecture works by constructing a recurrent model that predicts abstract internal states. These states are never directly related to the dynamics of the environment, but are learnt only through the signal coming from the total reward(s) obtained. Figure 1b shows a representation of this architecture.




(a) *Example of the abstractions obtained by the Predictron (Silver et al., 2017 [20]). Here the goal is to find the path connecting the two dark dots. The $g^k$, in order of increasing $k$, show the internal states learnt by the Predictron. As it can be seen, the learnt states solve the problem by sequentially clustering together (abstracting) the actions.*




(b) *The Predictron architecture (Silver et al., 2017 [20]). Here the different $s^k$ represent the internal abstract states. Each one of them makes a prediction for the reward and the value function corresponding to its state. They all also generate a $\lambda^k$ which is used to combine the results in a method similar to the $\lambda$-return in the TD($\lambda$) algorithm [23].*

### 2.2 Predictron for Temporally Abstract MDPs

Our objective is to use the Predictron to learn a temporally abstract MDP representing the original environment. This poses several challenges. Most importantly, because the original Predictron paper studies the problem in MRPs, it is able to reframe the problem so that to

provide more learning signal. However, in MDPs, the agent would only be able to access the *scalar* total reward. Another potential problem is that the actions of the abstract MDP are not known. This means that we need to either learn a model to predict what these actions are or model them directly into the world model, obtaining an abstract MRP instead. Both directions are interesting and we plan on exploring both.

### 2.3 Abstract States To Condition a Lower-Level Policy

We plan on using the learnt abstract states to condition a lower-level policy. Our intuition is the following. Model-free methods only have access to the scalar value function and ignore all the internal states that must have been implicitly learnt by the neural network to calculate it. This means that normal model-free methods don't access all of the information the value function contains. By conditioning the policy on the temporally-abstract internal states, we would provide it with this extra information. We believe these states would be more meaningful for planning than normal goals [15, 6] because, as noted by Schrittwieser et al. [18], the actions an agent takes are chosen to maximize the value function, meaning that latent states optimized to predict the value function should offer valuable information for planning.

### 2.4 Further Planned Improvements

We also plan to test whether other techniques from more traditional world models could be useful in our setup. These include using discrete state spaces [7], utilizing both deterministic and probabilistic hidden states to learn the dynamics of the world model [8] and using new sequence prediction methods like xLSTMs [2]. We also anticipate that one of the major problems our method will have to overcome is the overall weaker learning signal, especially problematic in sparse reward environments. We plan to mitigate this by partially grounding the internal states to predict the rewards that were actually obtained within the time period each of them covers (which can be calculated with the learnt discount factor $\gamma^k$ associated with each state). Of course, errors in the calculation of the time period of each state could compound and result in later states getting paired with the wrong rewards. We plan on utilizing techniques like reward smoothing [13] to mitigate this problem.

### 2.5 Related Works

Some other works also use architectures inspired by the Predictron to infer policies that act in an MDP [4, 16]. However, by grounding the model to work on a step-by-step basis, they focus on using it for planning without making use of the temporal abstraction potential that the Predictron architecture offers. Other works take drastically different approaches to deal with long term planning. The most common method is to couple model-based RL with hierarchical techniques [6, 15] to create short-term goals for the lower-level policy. However, goals often lie in the same latent space of either the world model or the policy, bounding them to represent only a specific instant in time. This means that to have more fine grained control over the lower-level policy, more intermediate goals are necessarily required, limiting long-term planning. Methods based on Successor Features [1] are also used since they can be thought of as using temporally extended actions [24], but are limited to value-based methods and require learning multiple different policies.

### 2.6 Limitations

The main limitation of the proposed method is that it learns the world model and value function jointly. This means that it relies on a specific task (rewards) to learn the world model, making

transfer learning to new tasks more challenging than it is with approaches using unsupervised exploration [19]. However, this also means that the world model of our proposed method will be tailored to the specific task at hand, which may help with overall performance.

## 3 Deliverables

The final project will take the form of a program implementing an agent able to learn a temporally abstract MDP as a world model of an environment (e.g. Minigrid [3], MiniHack [17]). The agent will also use this world model to augment its observations when choosing the actions to take. The final repository will also include tools to analyse the obtained internal states and understand which abstract states relate to which downstream actions.

## 4 Timeline

We plan to organize the project according to the following timeline.

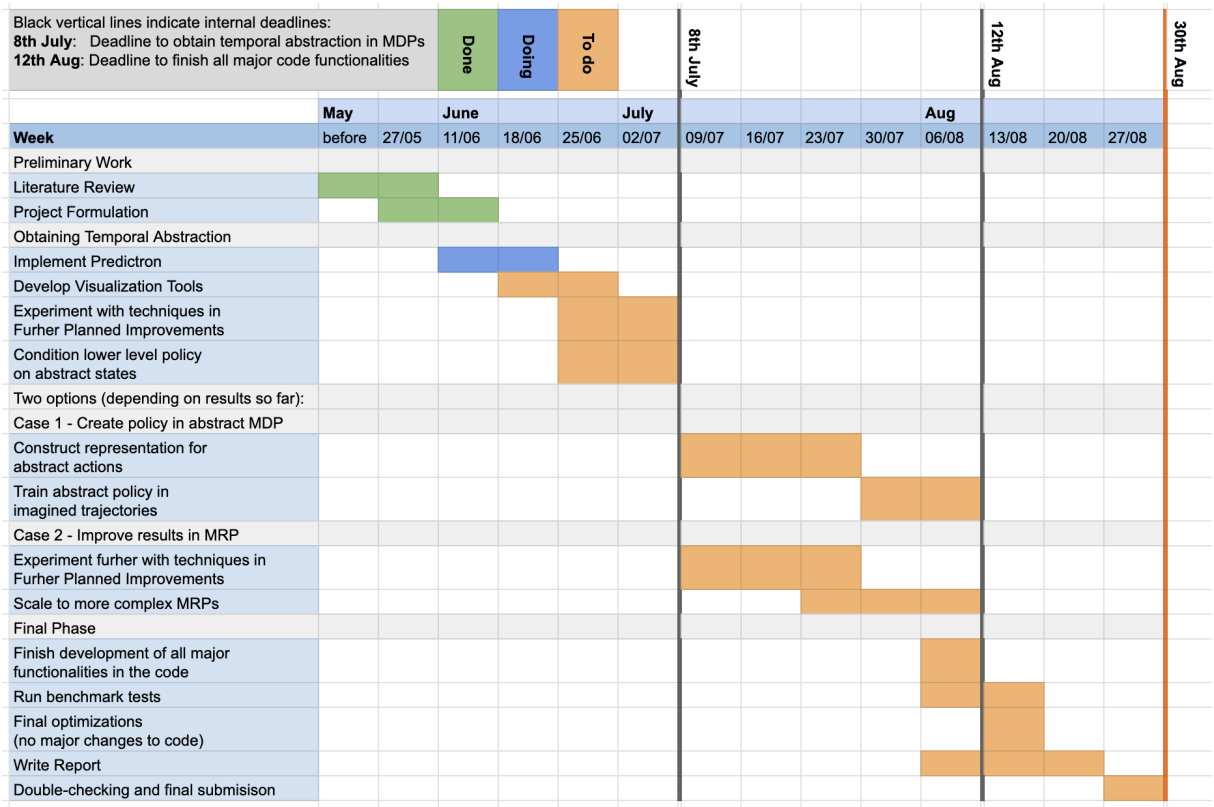



Figure 2: *Gantt chart for the project plan timeline.*

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